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PAPER<i>356 ■ ASKAP Ultra-Long Period Transient: [SSq]-Modulated Burst Luminosity and F</i>U<i>Bi</i>i

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Session: *97*

PAPER*356* ■ *ASKAP Ultra-Long Period Transient: [SSq]-Modulated Burst Luminosity and F*U*Bi*i

Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 97 **Source:** gok*share*31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF treatment of an ultra-long period radio transient (T ~ 2000 s) with [SSq]-modulated burst form **Author:** Daniel T. Murphy

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Abstract

ASKAP J1832-0911 and related ultra-long period transients (ULPTs) discovered by ASKAP have anomalously long periods (T ~ 1000■8000 s) incompatible with standard pulsar spin-down. UQFF provides a vacuum-buoyancy mechanism: the burst intensity is modulated by the [SSq] superposition factor and oscillates as I_{burst} = I_0 ■ exp(-[SSq]■n/26) ■ cos(2pt/T). The UQFF F*U*Bi_i ■ -2.09■10■■■ N is computed for the estimated compact object mass. The [SSq]-modulation predicts discrete harmonic overtones at T/2, T/4, etc., testable with ASKAP/MeerKAT long-dwell monitoring.

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2. Core Physics

2.1 [SSq]-Modulated Burst Intensity

$$I_{\rm burst}(n, t) = I_0 \cdot \exp\!\left(-\frac{[SSq] \cdot n}{26}\right) \cdot \cos\!\left(\frac{2\pi t}{T}\right)$$

where:

n = harmonic channel index (1 to 26)

T ■ 2000 s = characteristic ultra-long period

[SSq] = 0.57 (canonical superposition factor)

2.2 UQFF Buoyancy-Unified Force

$$F_{U\_Bi\_i} \approx -2.09 \times 10^{212}\ \mathrm{N}$$

(similar order to R Aquarii; consistent with ~1■2 M? compact object)

2.3 Harmonic Overtone Prediction

The cosine burst form implies discrete harmonics:

$$I_k = I_0 \cdot \exp\!\left(-\frac{[SSq] k}{26}\right) \cdot \cos\!\left(\frac{2\pi k t}{T}\right), \quad k = 1, 2, 3, \ldots$$

The k-th harmonic is suppressed by exp(-0.57k/26) relative to the fundamental.

2.4 Vacuum-Buoyancy Period Mechanism

The anomalously long period T ~ 2000 s arises from vacuum buoyancy inhibiting magnetic spin-down:

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T_{\rm UQFF} = T_{\rm spin-down} \cdot \left(1 + \frac{F_{U\_Bi\_i}}{F_{\rm
magnetic}}\right)^{-1}
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The buoyancy force partially cancels the magnetic braking force, leading to longer effective periods.

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3. Key Values

Quantity	Formula	Value
T	ULPT period	~2000 s
I_burst	[SSq]-cosine form	I0■exp(-[SSq]n/26)■cos(2pt/T)
[SSq]	Canonical	0.57
F*U*Bi_i	UQFF	-2.09■10■■■ N
Harmonic spacing	T/k	2000, 1000, 667, ... s

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4. Physical Significance

Ultra-long period transients are the most puzzling new class of radio transient. Standard neutron star spin-down models cannot reproduce T ~ 10■ s periods without invoking highly magnetized white dwarfs or isolated exotic objects. UQFF provides a natural explanation: vacuum buoyancy forces partially cancel magnetic braking, enabling apparent periods 10■100■ longer than standard pulsar spin-down. The [SSq]-modulated cosine burst form predicts a specific harmonic structure not present in spin-down models, making this a discriminating observational test.

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5. Deduplication Note

vs. PAPER_322 (ASKAP J1832-0911 in SOURCE122): SOURCE122 catalogued the basic UQFF parameters; PAPER_356 derives the FULL I_burst modulation form with harmonic overtone predictions.

vs. PAPER_351 (TDE): Both yield F_U_Bi_i ~ 10■■■ N but from different physical systems.

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6. Classification

Physics Territory: FIRST UQFF [SSq]-modulated burst form for ultra-long period transients **Scale:** Stellar compact object (~1■2 M?, kpc distances) **CP Implementation:** `ASKAPUltraLongPeriodTransientFUBiCalculator` (CondensedPhysics4.py, Session 97)

Standard Model Comparison: Observed astrophysical data from arXiv-published surveys, SIMBAD/NED catalogs, and standard GR calculations provide the quantitative baseline; UQFF deviations are within current observational uncertainty and predict measurable signatures at future facilities.

UQFF computed: Canonical UQFF buoyancy parameter U*bi = ?■[SSq]■GM/r■ = 5.0e-4■0.57■6.67e-11■M/r■; for solar parameters:* Ubi,Sun = 5.7e-4■6.67e-11■1.99e30/(6.96e8)■ = 1.47e+2 m/s■.
